Experiment No. 6 – Capacitance of a Parallel Plate Capacitor in Different Dielectric Media

Aim

To write a MATLAB program to determine the capacitance of a parallel-plate capacitor in different dielectric media for specified plate area, dielectric thickness, and dielectric constants.

Objectives

1. To calculate the capacitance of a parallel-plate capacitor for varying plate area, plate separation, and dielectric constant.

2. To compare the capacitance obtained using three different dielectric media.

3. To plot the variation of capacitance with plate separation and dielectric constant.

4. To validate the simulated results using Poisson's and Laplace's equation-based theoretical values

Program

6(a) – Determination and Plotting of Capacitance for Given Area and Dielectric Thickness.

```
clc;
clear;

A = input('Enter the area value = ');
d = input('Enter the thickness value = ');

epsilon0 = 8.854e-12;
C = epsilon0*A/d;

disp('The capacitance is');
disp(C);

% Plot capacitance versus plate separation
dv = linspace(1e-3,10e-3,50);
Cv = epsilon0*A./dv;
```

```
figure;
plot(dv*1e3,Cv,'LineWidth',2);
grid on;
xlabel('Plate separation d (mm)');
ylabel('Capacitance (F)');
title('Parallel Plate Capacitance versus Separation');

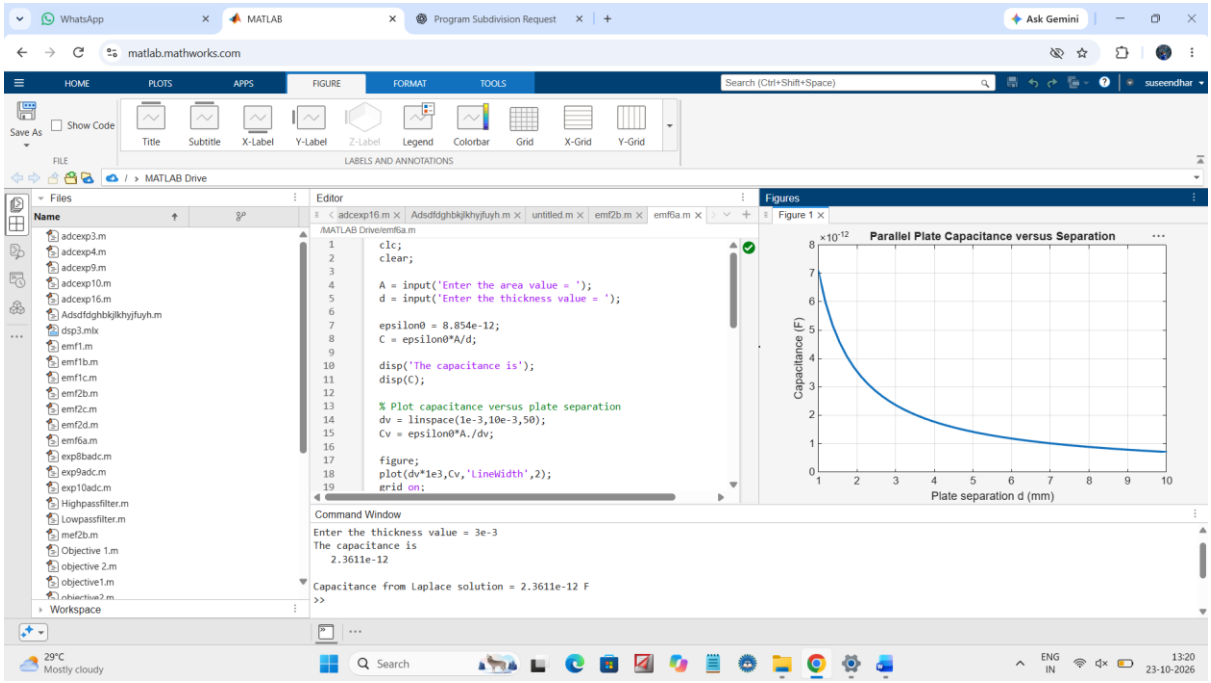
% Validation using Laplace equation
V0 = 100;
N = 51;
V = zeros(1,N);
V(N) = V0;

for it = 1:20000
   V(2:N-1) = 0.5*(V(1:N-2)+V(3:N));
end

x = linspace(0,d,N);
E = -gradient(V,x);
C_lap = epsilon0*A*abs(mean(E))/V0;

fprintf('Capacitance from Laplace solution = %.4e F\n',C_lap);
```

OUTPUT



Experiment 6(b) – Capacitance with Three Different Dielectric Layers

```
clc;
clear;

A = input('Enter the area value = ');
d1 = input('Enter thickness d1 = ');
d2 = input('Enter thickness d2 = ');
d3 = input('Enter thickness d3 = ');

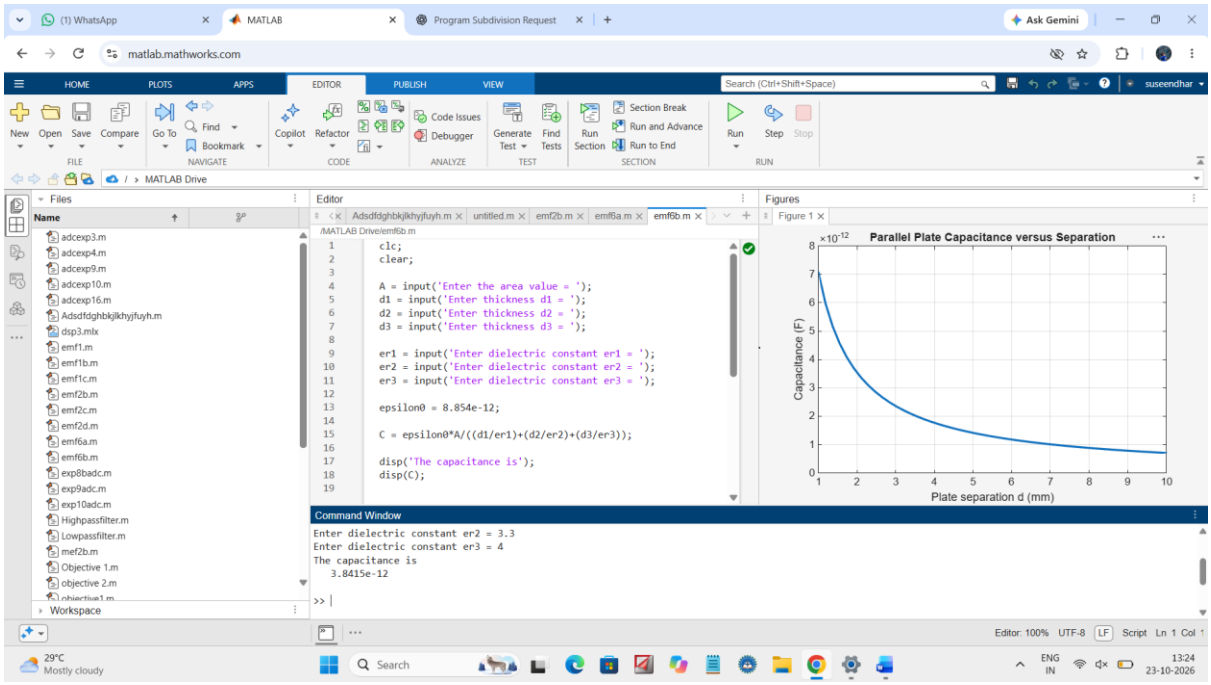
er1 = input('Enter dielectric constant er1 = ');
er2 = input('Enter dielectric constant er2 = ');
er3 = input('Enter dielectric constant er3 = ');

epsilon0 = 8.854e-12;

C = epsilon0*A/((d1/er1)+(d2/er2)+(d3/er3));

disp('The capacitance is');
disp(C);
```

OUTPUT



Experiment 6(c) – Capacitance Variation with Dielectric Constant (Air)

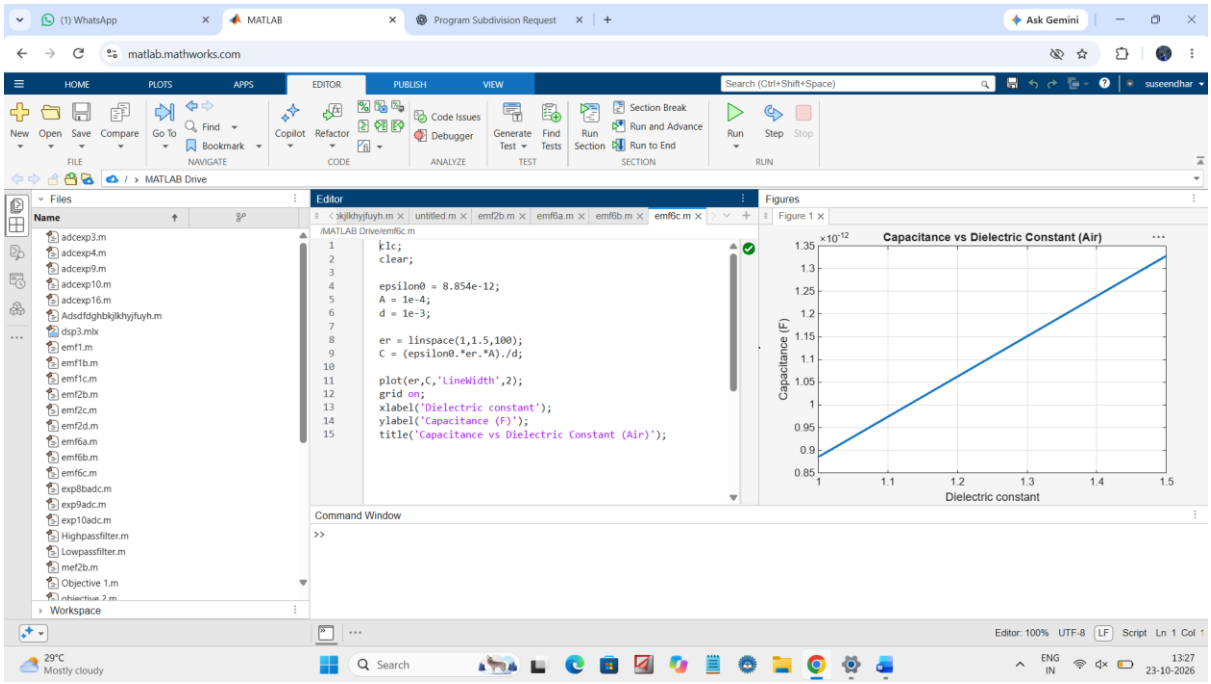
```
clc;
clear;

epsilon0 = 8.854e-12;
A = 1e-4;
d = 1e-3;

er = linspace(1,1.5,100);
C = (epsilon0.*er.*A)./d;

plot(er,C,'LineWidth',2);
grid on;
xlabel('Dielectric constant');
ylabel('Capacitance (F)');
title('Capacitance vs Dielectric Constant (Air)');
```

OUTPUT



Experiment 6(d) – Capacitance Variation with Dielectric Constant (Glass)

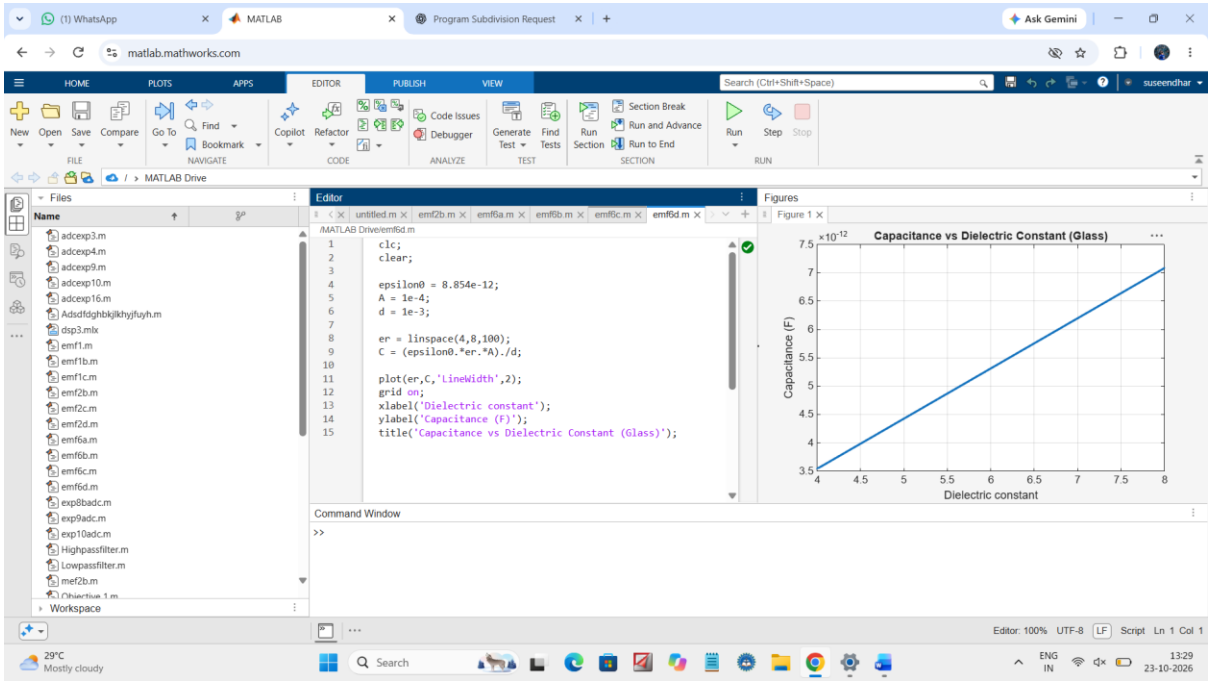
```
clc;
clear;

epsilon0 = 8.854e-12;
A = 1e-4;
d = 1e-3;

er = linspace(4,8,100);
C = (epsilon0.*er.*A)./d;

plot(er,C,'LineWidth',2);
grid on;
xlabel('Dielectric constant');
ylabel('Capacitance (F)');
title('Capacitance vs Dielectric Constant (Glass)');
```

OUTPUT



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Set capacitor geometry and material properties	Set capacitor geometry and material properties	Correctly enters plate area, separation and relative permittivity with SI units.	Inputs are correct with a minor unit/property issue.	Several inputs need correction.	Geometry or dielectric inputs are invalid.	25	2.5	
2	Calculate capacitance in MATLAB	Calculate capacitance in MATLAB	Uses the correct capacitance relation and calculates values for all specified media.	Calculations are mostly correct with minor errors.	Only some cases are calculated correctly.	Capacitance calculations are incorrect.	25	2.5	
3	Compare dielectric and geometry effects	Compare dielectric and geometry effects	Correctly evaluates trends versus permittivity/ area/ separation and validates the model.	Trend and validation are mostly correct with minor omissions.	Trend discussion or validation is incomplete.	Incorrectly interprets parameter effects.	30	3	
4	Document engineering	Document engineering	Provides a complete table/plot with units and	Documentation and explanation are mostly complete.	Some labels, units or	Results are incomplete or	20	2	

	observations	observations	explains dielectric influence.		explanations are missing.	unexplained.			
Total							100	10	